

Tracking Employment Changes in AI-Exposed Jobs

Bharat Chandar*

August 1, 2025

Abstract

This paper examines labor market trends for occupations exposed to Generative AI using US Current Population Survey (CPS) data from the post-ChatGPT era (Q4 2022 - Q1 2025). Contrary to displacement fears, occupations most exposed to AI on average show no substantial difference in employment or earnings growth compared to the least exposed. However, this aggregate trend masks heterogeneity across occupations. Within the most exposed quartile, occupations with a higher share of college-educated workers, such as software development, have experienced robust employment growth, while those with a lower share, like customer service, have seen declines. The analysis also reveals a divergence between job posting data and realized employment, urging caution when inferring labor market trends from postings alone.

1 Introduction

The period following the public release of ChatGPT in November 2022 has seen burgeoning research attempting to capture the labor market and productivity effects of generative AI, spurred by fear of potential widespread job displacement. This paper provides a timely empirical analysis of this evolving landscape by leveraging U.S. Current Population Survey

*Stanford University; chandarb@stanford.edu. I thank Erik Bynjolfsson, Basil Halperin, Philip Trammell, Luca Vendraminelli, Ruyu Chen, and Tom Mitchell for helpful comments.

measure AI’s impact. Recent findings have been varied. [Humlum and Vestergaard \(2025\)](#) use Danish administrative data to find minimal effects on earnings or hours worked, while [Jiang et al. \(2025\)](#) find AI exposure is correlated with longer work hours in the U.S.⁴ This paper adds additional color to this growing set of findings. In line with [Humlum and Vestergaard \(2025\)](#), I find little evidence of an aggregate decline in employment for exposed occupations. However, in contrast to their work, I find heterogeneity in employment trends across exposed occupations, with these differences correlated with features such as education. The divergent paths of software developers and customer service representatives, for example, suggest nuance in predictions of labor market changes for exposed workers.⁵

Finally, this work highlights an important methodological issue: the divergence between job postings and employment data. Recent studies have used job postings to measure shifts in labor demand from AI (e.g., [Acemoglu et al. \(2022\)](#); [Hampole et al. \(2025\)](#)). This paper shows that for key occupations, these two data sources tell different stories. The decline in software developer job postings since 2022, for instance, contrasts sharply with their stable-to-growing employment in the CPS. This finding serves as an important empirical caution in interpreting job posting trends as reflective of employment or earnings. The CPS data used here provides a transparent and timely alternative for tracking realized outcomes as they unfold for a representative sample of the U.S. population.

⁴See also [Acemoglu et al. \(2022\)](#); [Bonney et al. \(2024\)](#); [Bick et al. \(2024\)](#); [Hartley et al. \(2024\)](#); [Frank et al. \(2025\)](#); [Chen et al. \(2025\)](#); [Hampole et al. \(2025\)](#). The Federal Reserve Bank of New York tracks employment by college major using the American Community Survey (ACS), as documented in [Abel and Deitz \(2017\)](#). While the ACS has rich sociodemographic and economic information, it is released annually and with a time lag. As of Q2 2025 the latest ACS data is from 2023, in the initial year of LLM adoption.

⁵Another closely related paper is [Deming et al. \(2025\)](#), which combines data from the decennial Census, American Community Survey, and CPS to track labor market changes over 120 years. This paper complements their work by directly associating employment changes with measures of AI exposure and by tracking labor market trends specifically over the post-ChatGPT time period.

in months four and eight of the sample.⁶ Because the MORG data is a subset of the basic monthly CPS it has a smaller sample size. The MORG data underwent changes in topcoding conventions over the sample period. I manually topcode nominal weekly earnings at \$2884.61 per week to maintain consistency over time. I also convert nominal weekly earnings to real weekly earnings using the CPI-U.

I merge the CPS data with the AI exposure estimates using the occupation code, using a crosswalk from the BLS to merge 2018 SOC codes to Census occupation codes.

Computing employment trends by occupation can lead to noisy estimates due to small sample sizes. Further, the public monthly CPS file does not offer a straightforward method to compute standard errors for employment estimates by occupation. To mitigate these issues, I compute the average monthly employment in each quarter and consider only analyses with at least 100 observations in each cell.

3.3 Summary Statistics

Table 1 shows the most and least AI-exposed occupations based on the GPT-4-based β measure from Eloundou et al. (2024). Professions involving translation, writing, and programming are among the most exposed, while occupations involving manual labor and construction are among the least exposed.

4 Results

4.1 Labor Market Trends by AI Exposure

Across all quartiles of AI exposure, there is little evidence of job loss or slower earnings growth for more exposed occupations since the release of ChatGPT. Figure 1 shows the employment and earnings trends for each quartile of AI exposure measured using the GPT-

⁶Workers are surveyed for four months consecutively and then rotated out of the sample for four months before being rotated back in for another four months.

Table 1: Most and Least AI-Exposed Occupations. Exposure is defined using the GPT-4-based β measure from Eloundou et al. (2024). Occupations are 2018 Census occupation titles from the CPS merged via crosswalk to the 2018 SOC codes.

Most Exposed Occupations	Least Exposed Occupations
Mathematicians	Athletes and sports competitors
Court reporters and simultaneous captioners	Packaging and filling machine operators
Proofreaders and copy markers	Paper goods machine setters and operators
Correspondence clerks	Roustabouts, oil and gas
Computer programmers	Bus and truck mechanics
Web developers	Automotive glass installers and repairers
Database administrators and architects	Riggers
Telephone operators	Electrical power-line installers
Interpreters and translators	Excavating machine operators
Writers and authors	Rail-track maintenance operators

4-based β metric. The plots shows the average employment and average real usual weekly earnings in each quarter for each quartile of AI exposure. Each series is normalized to 1 in Q4 2022, corresponding to the time when ChatGPT was first released. The results suggest little if any evidence of relative employment declines in the most AI-exposed occupations since the release of ChatGPT. Real earnings have also grown at a similar rate. Figures A1 through A3 show similar results using alternative measures of AI exposure.

4.2 Software and Customer Service as Case Studies

Trends in the aggregate data mask the divergent patterns for key high-exposure occupations. This section presents two case studies—software developers and customer service representatives—that highlight this heterogeneity. Table 1 shows that software development occupations have some of the highest estimated exposure to AI, consistent with LLMs being relatively proficient in writing computer code. Figure 2 shows the employment trends for the Census occupation code corresponding to software developers. The results suggest an



Figure 1: Employment and earnings trends by AI exposure. The top panel shows the share of employment by quartile of AI exposure, normalized to start at 1 in 2022 Q4. The AI exposure measure is the GPT-4-based β measure from Eloundou et al. (2024). Earnings are real usual weekly earnings, adjusted for inflation using the CPI-U and normalized to start at 1 in 2022 Q4.